

ADITYA VATS

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PROFILE SUMMARY

Final-year B.Tech CSE student with strong hands-on experience in **Data Analysis** and applied **Machine Learning**. Proficient in **Python** and **SQL**, with experience in EDA and building predictive models for real-world datasets. Worked on large-scale transactional and **NLP**-based projects involving classification and similarity matching. Seeking a **Data Analyst** or **Data Science Internship** to apply analytical skills in data-driven decision-making.

EDUCATION

B.Tech, Computer Science & Engineering, VIT-AP University
CGPA: 8.06 / 10

Aug 2022 – May 2026

SKILLS

Data Analysis & Analytics:	Python (Pandas, NumPy), SQL (Joins, Aggregations), EDA, Data Cleaning, Feature Engineering, Matplotlib, Seaborn
Machine Learning (Applied):	Regression, Classification, Random Forest, XGBoost, Model Evaluation (F1, Recall, ROC-AUC), Handling Imbalanced Data, NLP
Tools & Backend Exposure:	FastAPI, REST APIs, Git, GitHub, Docker (basic), AWS EC2
Databases:	MySQL, MongoDB
CS Fundamentals:	DSA, OOP Concepts

EXPERIENCE

Software / Data Intern— Blackbucks Education

Feb 2025 – Jun 2025

- Built backend REST APIs for a production MERN application; reduced API latency by **30%**. Collaborated with frontend and database layers to **integrate APIs**, handle authentication flows, and improve overall application.

PROJECTS

Fraud Detection System (Capstone Project): Developed an **end-to-end** financial fraud detection system on a dataset of over **6M+ transactions**, incorporating classical ML, deep learning, unsupervised and quantum models with REST API deployment via FastAPI. Evaluated multiple models including XGBoost (**99.2% accuracy, 94% F1**), MLP (**98.8% accuracy, 93.5% F1**), and Autoencoder anomaly detection (**AUC 0.993, F1 0.901**), optimizing for real-world imbalance and detection performance while balancing accuracy and recall priorities.

Resume-Job Description Matching System (NLP): Built an **NLP-based resume screening system** using text preprocessing and **TF-IDF vectorization** to represent resumes and job descriptions numerically. Applied **cosine similarity** to rank resumes by relevance, enabling automated and scalable resume-JD matching for candidate shortlisting.

Image Captioning using CNN and RNN: Developed an end-to-end image captioning model combining **DenseNet201** for visual feature extraction and **LSTM** for sequence modeling. Trained on **40,000+ images** using transfer learning techniques and improved caption coherence through data preprocessing and hyperparameter tuning.

CERTIFICATIONS

- Deloitte Australia Data Analytics Job Simulation (Forage)** — Built Tableau **dashboards** for factory downtime analysis and performed forensic pay-equity **classification**.
- AWS Academy Cloud Foundations** — Covered AWS fundamentals including EC2, S3, IAM, networking, and cloud security basics.
- AWS Academy Cloud Architecting** — Learned to design and deploy scalable, cost-efficient AWS solutions using EC2, RDS, and load balancing.